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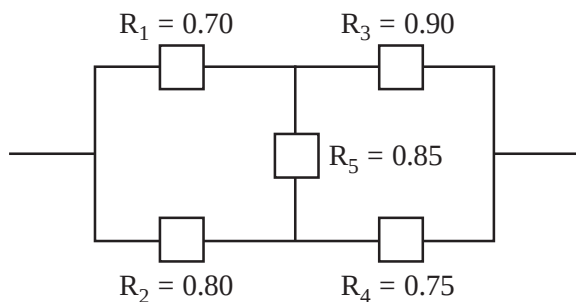
Roll No

EE-8402 (GS)
B.E. VIII Semester Examination, June 2020
Grading System (GS)
Power System Planning and Reliability
(Elective-IV)
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. Define following terms :
 - i) Random variable
 - ii) PMF
 - iii) PDF
 - iv) CDF
2. What are Binomial and Poission distributions? What conditions must an experiment or event fulfill if Binomial or Poission distribution is to be used?
3. A stockpile of 40 relays contain 8 defective relay. If five relays are selected at random and the number of defective relays in known to be greater than two, what is the probability that exactly four relays are defective?
4. Define the following terms
 - a) Minimal tie-set
 - b) Minimal cut set
 - c) Tie-set
 - d) Cut-set
5. Evaluate the reliability of the following system using the decomposition method.



6. How will you determine the reliability of series and parallel systems?
7. Compare correlated and uncorrelated load models.

OR

What are the effects of varying load in bulk power system?

8. Write short notes on any two of the following:
 - a) Effect of varying load on composite system reliability
 - b) Merging of generation and load models
 - c) Switching after faults
 - d) Cost and Worth of reliability
